

We will cover the following five topics in the Week 8 lectures:

- 1 Plane transformations and space transformations. *generally non-linear determinant $\Rightarrow$ max 3×3 matrix*
- 2 The Jacobian of a plane/space transformation.
- 3 Change of variables in multivariable integration. *u-substitution in higher dimension*
- 4 Line integrals of a scalar/vector field. *vector calculus.*

Midterm Exam 2

- 1: Lagrange Multipliers & maximization/minimization problem with boundary.
- 2: Double/Triple integrals (different types, cylindrical/spherical coord)
- 3: Change of variables & line integrals.

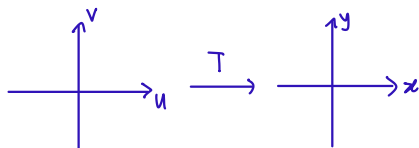
The Geometrical Meaning of the Determinant

In Linear Algebra I, you will have learned about the determinant.

Let T be a linear transformation from the (u, v) -plane to the (x, y) -plane.

This means that we can find $a, b, c, d \in \mathbb{R}$ such that

$$\forall u, v \in \mathbb{R} : T(u, v) = u \cdot (a, b) + v \cdot (c, d) = (au + cv, bu + dv).$$



$$T\left(\begin{bmatrix} u \\ v \end{bmatrix}\right) = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}$$

$$T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} c \\ d \end{bmatrix}$$

The Geometrical Meaning of the Determinant

Let S denote the closed standard square in the (u, v) -plane with vertices

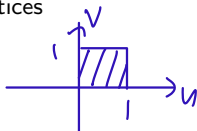
$(0, 0)$, $(1, 0)$, $(1, 1)$, and $(0, 1)$.

Linear Transformation respects addition

Observe that

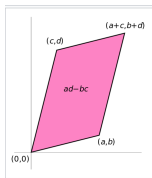
$$T((1, 0) + (0, 1)) = T(1, 0) + T(0, 1)$$

$$T(1, 0) = (a, b), \quad T(1, 1) = (a + c, b + d), \quad \text{and} \quad T(0, 1) = (c, d).$$



Hence, T turns S into the closed parallelogram P in the (x, y) -plane with vertices

$(0, 0)$, (a, b) , $(a + c, b + d)$, and (c, d) .



det = 0 $\Rightarrow$ no area

$$\text{Note that } \text{Area}(P) = |ad - bc| = \left| \det \left(\begin{bmatrix} a & c \\ b & d \end{bmatrix} \right) \right| = |\det(T)|.$$

area of parallelogram
 $\rightarrow$ geometrical interpretation

The Geometrical Meaning of the Determinant

Let R be a closed standard rectangle in the (u, v) -plane.

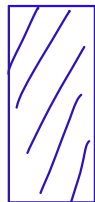
Suppose that the u - and v -sides of R have lengths Δu and Δv , respectively.

Then T turns R into a closed parallelogram in the (x, y) -plane with area

$$|\det(T)| \cdot (\Delta u)(\Delta v).$$

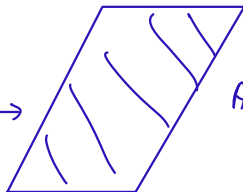
This suggests that $dA(x, y) = |\det(T)| \cdot dA(u, v)$.

u, v plane



Δv

$$\text{area} = (\Delta u)(\Delta v)$$



$$\text{Area} = |\det(T)| \cdot (\Delta u)(\Delta v)$$

Δu

The Geometrical Meaning of the Determinant

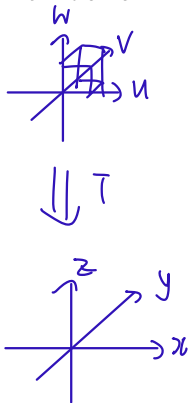
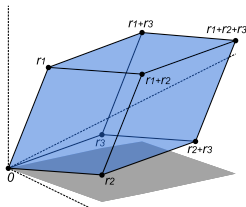
Let T now be a linear transformation from (u, v, w) -space to (x, y, z) -space.

Let B be a closed standard box in (u, v, w) -space.

Suppose that the u -, v -, and w -sides of B have lengths Δu , Δv , and Δw , respectively.

Then T turns B into a closed parallelepiped in (x, y, z) -space with volume

$$|\det(T)| \cdot (\Delta u)(\Delta v)(\Delta w).$$



This suggests that $dV(x, y, z) = |\det(T)| \cdot dV(u, v, w)$.

volume in
(x, y, z) space volume in
(u, v, w) space

The Geometrical Meaning of the Determinant

Theorem (Transformation of a Double Integral Under a Linear Transformation)

- Let D be a bounded region in the (u, v) -plane whose area exists.
- Let T be a linear transformation from the (u, v) -plane to the (x, y) -plane.
- Let f be a real-valued function of two real variables such that $T[D] \subseteq \text{Dom}(f)$.

Remark: $T[D]$ denotes the image of D under T and is a bounded region in $\mathbb{R}^2$.

If $\iint_{T[D]} f(x, y) \, dA(x, y)$ exists, then so does

↓
linear transformation

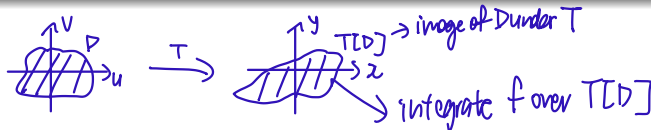
$$\iint_D f(T(u, v)) \cdot |\det(T)| \, dA(u, v),$$

in which case

Integral in (x, y) plane

integral in (u, v) plane

$$\iint_{T[D]} f(x, y) \, dA(x, y) = \iint_D f(T(u, v)) \cdot \overbrace{|\det(T)|}^{\text{number}} \, dA(u, v).$$



The Geometrical Meaning of the Determinant

Theorem (Transformation of a Triple Integral Under a Linear Transformation)

- Let R be a bounded region in (u, v, w) -space whose volume exists.
- Let T be a linear transformation from (u, v, w) -space to (x, y, z) -space.
- Let f be a real-valued function of three real variables such that $T[R] \subseteq \text{Dom}(f)$.

Remark: $T[R]$ denotes the image of R under T and is a bounded region in $\mathbb{R}^3$.

If $\iiint_{T[R]} f(x, y, z) \, dV(x, y, z)$ exists, then so does

$$\iiint_R f(T(u, v, w)) \cdot |\det(T)| \, dV(u, v, w),$$

in which case

triple integral in (x, y, z) space *Triple integral in (u, v, w) space*

$$\iiint_{T[R]} f(x, y, z) \, dV(x, y, z) = \iiint_R f(T(u, v, w)) \cdot \overbrace{|\det(T)|}^{\text{number}} \, dV(u, v, w).$$

integrate f over $T[R]$

The Geometrical Meaning of the Determinant

Corollary (Area Change Under a Linear Transformation)

- Let D be a bounded region in the (u, v) -plane whose area exists.
- Let T be a linear transformation from the (u, v) -plane to the (x, y) -plane.

Then the area of $T[D]$ exists, and $\text{Area}(T[D]) = |\det(T)| \cdot \text{Area}(D)$.

Corollary (Volume Change Under a Linear Transformation)

- Let D be a bounded region in (u, v, w) -space whose volume exists.
- Let T be a linear transformation from (u, v, w) -space to (x, y, z) -space.

Then the volume of $T[D]$ exists, and $\text{Vol}(T[D]) = |\det(T)| \cdot \text{Vol}(D)$.

$$\text{new area/volume} = |\det(T)| \cdot \text{old area/volume}$$

Plane Transformations

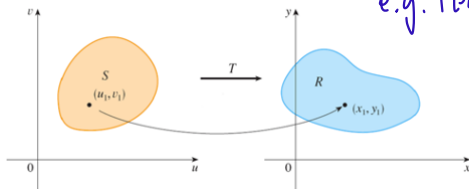
→ func. from subset of $\mathbb{R}^2$ to subset of $\mathbb{R}^2$
↳ doesn't have to be linear!

Definition (Plane Transformations)

A **plane transformation** is a function T with the following two properties:

- $\text{Dom}(T)$ is a subset of the (u, v) -plane.
- $\text{Range}(T)$ is a subset of the (x, y) -plane.

Remark: T has two scalar component functions, X and Y , so $T = (X, Y)$.



e.g. $T(x, y) = (xe^y, x + y^2)$

Example

All linear transformations from the (u, v) -plane to the (x, y) -plane.

→ func. from subset of $\mathbb{R}^2$ to subset of $\mathbb{R}^3$.

Definition (Space Transformations)

A **space transformation** is a function T with the following two properties:

- $\text{Dom}(T)$ is a subset of (u, v, w) -space.
- $\text{Range}(T)$ is a subset of (x, y, z) -space.

Remark: T has three scalar component functions, X , Y , and Z , so $T = (X, Y, Z)$.

Example

All linear transformations from (u, v, w) -space to (x, y, z) -space.

The Jacobian of a Plane Transformation

That's 2 outputs. $\Rightarrow X, Y$ are the components of T .

Definition (Jacobian of a Plane Transformation)

Let $T = (X, Y)$ be a plane transformation from the (u, v) -plane to the (x, y) -plane.

The **Jacobian of T** , denoted by J_T , is a real-valued function defined as follows:

$$J_T \stackrel{\text{df}}{=} \det \left(\begin{bmatrix} \frac{\partial X}{\partial u} & \frac{\partial X}{\partial v} \\ \frac{\partial Y}{\partial u} & \frac{\partial Y}{\partial v} \end{bmatrix} \right).$$

Theorem (Jacobian of a Linear Plane Transformation)

For every linear transformation from the (u, v) -plane to the (x, y) -plane,

$$J_T \equiv \det(T).$$

The Jacobian of a Plane Transformation — Polar Coordinates

Define a plane transformation, T_{polar} , from the (r, θ) -plane to the (x, y) -plane by

$$T_{\text{polar}}(r, \theta) \stackrel{\text{df}}{=} (r \cos(\theta), r \sin(\theta)).$$

Given $r, \theta \in \mathbb{R}$, compute $J_{T_{\text{polar}}}(r, \theta)$ as follows:

$$\begin{aligned} \det \begin{pmatrix} \frac{\partial}{\partial r} [r \cos(\theta)] & \frac{\partial}{\partial \theta} [r \cos(\theta)] \\ \frac{\partial}{\partial r} [r \sin(\theta)] & \frac{\partial}{\partial \theta} [r \sin(\theta)] \end{pmatrix} &= \det \begin{pmatrix} \cos(\theta) & -r \sin(\theta) \\ \sin(\theta) & r \cos(\theta) \end{pmatrix} \\ &= r \cos^2(\theta) + r \sin^2(\theta) \\ &= r. \end{aligned}$$

The Jacobian of a Space Transformation

Definition (Jacobian of a Space Transformation)

Let $T = (X, Y, Z)$ be a space transformation from (u, v, w) -space to (x, y, z) -space.

3 components

The **Jacobian of T** , denoted by J_T , is a real-valued function defined as follows:

T has 3 output

$$J_T \stackrel{\text{df}}{=} \det \begin{pmatrix} \frac{\partial X}{\partial u} & \frac{\partial X}{\partial v} & \frac{\partial X}{\partial w} \\ \frac{\partial Y}{\partial u} & \frac{\partial Y}{\partial v} & \frac{\partial Y}{\partial w} \\ \frac{\partial Z}{\partial u} & \frac{\partial Z}{\partial v} & \frac{\partial Z}{\partial w} \end{pmatrix} = \frac{\partial(x, y, z)}{\partial(u, v, w)} \quad \text{same thing.}$$

Theorem (Jacobian of a Linear Space Transformation)

For every linear transformation from (u, v, w) -space to (x, y, z) -space,

$$J_T \equiv \det(T).$$

The Jacobian of a Space Transformation — Cylindrical Coordinates

Define a space transformation, $T_{\text{cylindrical}}$, from (r, θ, z) -space to (x, y, z) -space by

$$T_{\text{cylindrical}}(r, \theta, z) \stackrel{\text{df}}{=} (r \overset{x}{\cos(\theta)}, r \overset{y}{\sin(\theta)}, \overset{z}{z}).$$

Given $r, \theta, z \in \mathbb{R}$, compute $J_{T_{\text{cylindrical}}}(r, \theta, z)$ as follows:

$$\begin{aligned} & \det \begin{pmatrix} \frac{\partial}{\partial r} [r \overset{x}{\cos(\theta)}] & \frac{\partial}{\partial \theta} [r \overset{x}{\cos(\theta)}] & \frac{\partial}{\partial z} [r \overset{x}{\cos(\theta)}] \\ \frac{\partial}{\partial r} [r \overset{y}{\sin(\theta)}] & \frac{\partial}{\partial \theta} [r \overset{y}{\sin(\theta)}] & \frac{\partial}{\partial z} [r \overset{y}{\sin(\theta)}] \\ \frac{\partial}{\partial r} [\overset{z}{z}] & \frac{\partial}{\partial \theta} [\overset{z}{z}] & \frac{\partial}{\partial z} [\overset{z}{z}] \end{pmatrix} \\ &= \det \begin{pmatrix} \cos(\theta) & -r \sin(\theta) & 0 \\ \sin(\theta) & r \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{block matrix} \\ &= r \cos^2(\theta) + r \sin^2(\theta) \\ &= r. \end{aligned}$$

The Jacobian of a Space Transformation — Spherical Coordinates

Define a space transformation, $T_{\text{spherical}}$, from (ρ, θ, ϕ) -space to (x, y, z) -space by

$$T_{\text{spherical}}(\rho, \theta, \phi) \stackrel{\text{df}}{=} (\rho \sin(\phi) \cos(\theta), \rho \sin(\phi) \sin(\theta), \rho \cos(\phi)).$$

Given $\rho, \theta, \phi \in \mathbb{R}$, compute $J_{T_{\text{spherical}}}(\rho, \theta, \phi)$ as follows:

$$\begin{aligned} &= \det \left(\begin{bmatrix} \frac{\partial}{\partial \rho} [\rho \sin(\phi) \cos(\theta)] & \frac{\partial}{\partial \theta} [\rho \sin(\phi) \cos(\theta)] & \frac{\partial}{\partial \phi} [\rho \sin(\phi) \cos(\theta)] \\ \frac{\partial}{\partial \rho} [\rho \sin(\phi) \sin(\theta)] & \frac{\partial}{\partial \theta} [\rho \sin(\phi) \sin(\theta)] & \frac{\partial}{\partial \phi} [\rho \sin(\phi) \sin(\theta)] \\ \frac{\partial}{\partial \rho} [\rho \cos(\phi)] & \frac{\partial}{\partial \theta} [\rho \cos(\phi)] & \frac{\partial}{\partial \phi} [\rho \cos(\phi)] \end{bmatrix} \right) \\ &= \det \left(\begin{bmatrix} \sin(\phi) \cos(\theta) & -\rho \sin(\phi) \sin(\theta) & \rho \cos(\phi) \cos(\theta) \\ \sin(\phi) \sin(\theta) & \rho \sin(\phi) \cos(\theta) & \rho \cos(\phi) \sin(\theta) \\ \cos(\phi) & 0 & -\rho \sin(\phi) \end{bmatrix} \right). \end{aligned}$$

Take advantage of "0" for determinant calculation

The Jacobian of a Space Transformation — Spherical Coordinates (cont'd)

A co-factor expansion along the third row then yields

$$\begin{aligned} & \cos(\phi) \cdot \det \left(\begin{bmatrix} -\rho \sin(\phi) \sin(\theta) & \rho \cos(\phi) \cos(\theta) \\ \rho \sin(\phi) \cos(\theta) & \rho \cos(\phi) \sin(\theta) \end{bmatrix} \right) - \\ & \rho \sin(\phi) \cdot \det \left(\begin{bmatrix} \sin(\phi) \cos(\theta) & -\rho \sin(\phi) \sin(\theta) \\ \sin(\phi) \sin(\theta) & \rho \sin(\phi) \cos(\theta) \end{bmatrix} \right) \\ &= \cos(\phi) \cdot \left[-\rho^2 \sin(\phi) \cos(\phi) \sin^2(\theta) - \rho^2 \sin(\phi) \cos(\phi) \cos^2(\theta) \right] - \\ & \quad \rho \sin(\phi) \cdot \left[\rho \sin^2(\phi) \cos^2(\theta) + \rho \sin^2(\phi) \sin^2(\theta) \right] \\ &= -\rho^2 \sin(\phi) \cos^2(\phi) - \rho^2 \sin(\phi) \sin^2(\phi) \\ &= \textcircled{-\rho^2 \sin(\phi)}. \end{aligned}$$



we multiply
by absolute value
of Jacobian
in the previous week's formula

Change of Variables in Single-Variable Integration

Calculus II

Recall the following theorem for change of variables in single-variable integration.

Theorem (Change of Variables in Single-Variable Integration)

- Let a and b be real numbers such that $a < b$.
- Let g be a differentiable real-valued function of one real variable defined on $[a, b]$.
- Let f be a real-valued function of one real variable such that $g[[a, b]] \subseteq \text{Dom}(f)$.

If $\int_{g(a)}^{g(b)} f(u) \, du$ exists, then $\int_a^b f(g(x)) \cdot g'(x) \, dx$ also exists, and

$$\int_a^b f(g(x)) \cdot g'(x) \, dx = \int_{g(a)}^{g(b)} f(u) \, du.$$

↓
the image in $[a, b]$
under g is in
domain of f .

This theorem is often called the " **u -Substitution Method**": if we let $u = g(x)$, then

x -values

$$\int_a^b f(g(x)) \cdot g'(x) \, dx = \int_a^b f(u) \cdot \frac{du}{dx} \, dx = \int_{g(a)}^{g(b)} f(u) \, du.$$

u -values

$\frac{du}{dx} = g'(x)$

We now present generalizations of this result for two- and three-variable integration.

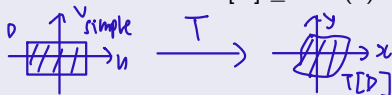
Change of Variables in Multivariable Integration

Tested!

Theorem (Change of Variables in Two-Variable Integration)

- Let D be a bounded region in the (u, v) -plane whose area exists.
- Let T be a plane transformation from the (u, v) -plane to the (x, y) -plane having the following four properties:
 - $\text{Dom}(T)$ is an open subset of the (u, v) -plane.
 - The entries of the Jacobian matrix of T are continuous throughout $\text{Dom}(T)$.
 - $D \subseteq \text{Dom}(T)$, and $T|_{D^\circ}$ (the restriction of T to the interior D° of D) is one-to-one.
 - The entries of the Jacobian matrix of $(T|_{D^\circ})^{-1}$ are continuous throughout $T[D^\circ]$.
- Let f be a real-valued function of two real variables such that $T[D] \subseteq \text{Dom}(f)$.

If $\iint_{T[D]} f(x, y) \, dA(x, y)$ exists, then so does



$$\iint_D f(T(u, v)) \cdot |J_T(u, v)| \, dA(u, v),$$

in which case

$$\iint_{T[D]} f(x, y) \, dA(x, y) = \iint_D f(T(u, v)) \cdot |J_T(u, v)| \, dA(u, v).$$

(x, y) integral (u, v) integral
 complicated simple in absolute values!



Change of Variables in Multivariable Integration

Theorem (Change of Variables in Three-Variable Integration)

- Let R be a bounded region in (u, v, w) -space whose volume exists.
- Let T be a space transformation from (u, v, w) -space to (x, y, z) -space satisfying the following four properties:
 - $\text{Dom}(T)$ is an open subset of (u, v, w) -space.
 - The entries of the Jacobian matrix of T are continuous throughout $\text{Dom}(T)$.
 - $R \subseteq \text{Dom}(T)$, and $T|_{R^\circ}$ (the restriction of T to the interior R° of R) is one-to-one.
 - The entries of the Jacobian matrix of $(T|_{R^\circ})^{-1}$ are continuous throughout $T[R^\circ]$.
- Let f be a real-valued function of three real variables such that $T[R] \subseteq \text{Dom}(f)$.

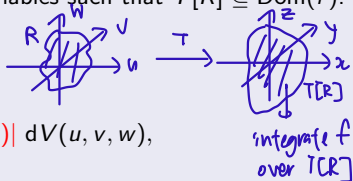
If $\iiint_{T[R]} f(x, y, z) \, dV(x, y, z)$ exists, then so does

$$\iiint_R f(T(u, v, w)) \cdot |J_T(u, v, w)| \, dV(u, v, w),$$

in which case

$$\iiint_{T[R]} f(x, y, z) \, dV(x, y, z) = \iiint_R f(T(u, v, w)) \cdot |J_T(u, v, w)| \, dV(u, v, w).$$

Handwritten notes: The region $T[R]$ in the first integral is circled and labeled "complicated". The region R in the second integral is circled and labeled "simple".



Change of Variables in Multivariable Integration — Polar Coordinates

Let D be a bounded region in the (r, θ) -plane whose area exists, such that $D \subseteq \mathbb{R}_{\geq 0}^r \times [0, 2\pi]^\theta$.
 $\nearrow \geq 0, \therefore r$ is the radial dist

Then T_{polar} is one-to-one within the interior of D . (Why?)

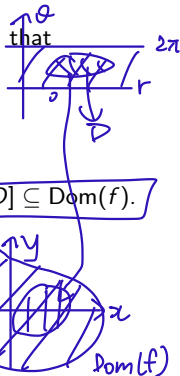
Let f be a real-valued function of two real variables such that $T_{\text{polar}}[D] \subseteq \text{Dom}(f)$.
 $\mathbb{R}_{>0} \times]0, 2\pi[\Rightarrow$ open interval

If $\iint_{T_{\text{polar}}[D]} f(x, y) \, dA(x, y)$ exists, then so does $\iint_D f(r \cos(\theta), r \sin(\theta)) \cdot r \, dA(r, \theta)$.
 ensures this makes sense

$$\iint_D f(r \cos(\theta), r \sin(\theta)) \cdot r \, dA(r, \theta),$$

in which case

$$\iint_{T_{\text{polar}}[D]} f(x, y) \, dA(x, y) = \iint_D f(\underbrace{r \cos(\theta)}_x, \underbrace{r \sin(\theta)}_y) \cdot \underbrace{r}_{J} \, dA(r, \theta).$$



Change of Variables in Multivariable Integration — Cylindrical Coordinates

Let R be a bounded region in (r, θ, z) -space whose volume exists, such that

$$R \subseteq \mathbb{R}_{\geq 0}^r \times [0, 2\pi]^\theta \times \mathbb{R}^z.$$

Then $T_{\text{cylindrical}}$ is one-to-one within the interior of R . (Why?)

Let f be a real-valued function of three real variables such that $T_{\text{cylindrical}}[R] \subseteq \text{Dom}(f)$.
Handwritten notes: $\subseteq \mathbb{R}_+ \times]0, 2\pi[, \mathbb{R}$ ensures that

If $\iiint_{T_{\text{cylindrical}}[R]} f(x, y, z) \, dV(x, y, z)$ exists, then so does *this triple integral makes sense*

$$\iiint_R f(r \cos(\theta), r \sin(\theta), z) \cdot r \, dV(r, \theta, z),$$

in which case

$$\iiint_{T_{\text{cylindrical}}[R]} f(x, y, z) \, dV(x, y, z) = \iiint_R f(\underbrace{r \cos(\theta)}_x, \underbrace{r \sin(\theta)}_y, \underbrace{z}_z) \cdot r \, dV(r, \theta, z).$$

tends to be complicated in (x, y, z) space

tends to be simple in (r, θ, z) space

Change of Variables in Multivariable Integration — Spherical Coordinates

Let R be a bounded region in (ρ, θ, ϕ) -space whose volume exists, such that

$$R \subseteq \overset{\rho}{\mathbb{R}_{\geq 0}} \times \overset{\theta}{[0, 2\pi]} \times \overset{\phi \rightarrow \text{inclination}}{[0, \pi]}.$$

Then $T_{\text{spherical}}$ is one-to-one within the interior of R . (Why?)

Let f be a real-valued function of three real variables such that $T_{\text{spherical}}[R] \subseteq \text{Dom}(f)$
 $\subseteq \mathbb{R}_{\geq 0} \times]0, 2\pi[\times]0, \pi[.$
If $\iiint_{T_{\text{spherical}}[R]} f(x, y, z) \, dV(x, y, z)$ exists, then so does $\iiint_R f(\rho \sin(\phi) \cos(\theta), \rho \sin(\phi) \sin(\theta), \rho \cos(\phi)) \cdot \rho^2 \sin(\phi) \, dV(\rho, \theta, \phi)$.
Ensures that this triple integral makes sense

$$\iiint_R f(\rho \sin(\phi) \cos(\theta), \rho \sin(\phi) \sin(\theta), \rho \cos(\phi)) \cdot \rho^2 \sin(\phi) \, dV(\rho, \theta, \phi),$$

in which case

$$\begin{aligned} & \iiint_{T_{\text{spherical}}[R]} f(x, y, z) \, dV(x, y, z) \\ &= \iiint_R f(\underbrace{\rho \sin(\phi) \cos(\theta)}_x, \underbrace{\rho \sin(\phi) \sin(\theta)}_y, \underbrace{\rho \cos(\phi)}_z) \cdot \rho^2 \sin(\phi) \, dV(\rho, \theta, \phi). \end{aligned}$$

Change of Variables in Multivariable Integration — Problem Solving

~~Difficult~~

In most cases, you will be given a complicated region of integration in the (x, y) -plane (or (x, y, z) -space).

The hope is that this region of integration is the image of a less complicated region in the (u, v) -plane (or (u, v, w) -space) under a plane (or space) transformation T .

The trouble is, we are usually given T^{-1} (the inverse of T) instead of T itself.

T^{-1} is a plane (or space) transformation from the (x, y) -plane (or (x, y, z) -space) to the (u, v) -plane (or (u, v, w) -space).

Given T^{-1} , we can usually solve for T itself, but the work can get quite messy.

Change of Variables in Multivariable Integration — Problem Solving

$T(x, y)$
Strategy for solving a change-of-variables problem in two-variable integration:

- Write $T^{-1} = (U, V)$.

backward transformation

- Compute the Jacobian $J_{T^{-1}} = \det$

$$\begin{bmatrix} \frac{\partial U}{\partial x} & \frac{\partial U}{\partial y} \\ \frac{\partial V}{\partial x} & \frac{\partial V}{\partial y} \end{bmatrix}$$

Expression in terms of variables x and y .

- Observe that

$$\iint_{T[D]} f(x, y) dA(x, y) = \iint_{T[D]} \frac{f(x, y)}{|J_{T^{-1}}(x, y)|} |J_{T^{-1}}(x, y)| dA(x, y).$$

- Letting $(u, v) = T^{-1}(x, y)$, write $\frac{f(x, y)}{|J_{T^{-1}}(x, y)|}$ *solely in terms of u and v* , i.e., find a real-valued function g of two real variables such that

$$\frac{f(x, y)}{|J_{T^{-1}}(x, y)|} = g(u, v).$$

doesn't necessarily require you to solve x and y in terms of u and v .

- Find the source region D in the (u, v) -plane (using figures if necessary).
- Then

$$\iint_{T[D]} f(x, y) dA(x, y) = \underbrace{\iint_D g(u, v) dA(u, v)}_{\text{Evaluate!}}$$

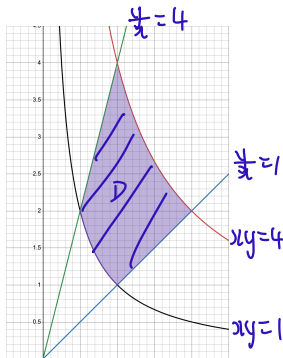
Change of Variables in Multivariable Integration — Example 1

Example

Let $D \stackrel{\text{df}}{=} \left\{ (x, y) \in \overbrace{\mathbb{R}_{>0} \times \mathbb{R}_{>0}}^{\text{first quadrant}} \mid 1 \leq \underbrace{xy}_u \leq 4 \text{ and } 1 \leq \underbrace{\frac{y}{x}}_v \leq 4 \right\}.$

Problem: Evaluate $\iint_D (x^2 + y^2) dA(x, y).$

Solution



• Let $u = xy, v = \frac{y}{x}$ } restrict to $(xy) \in \mathbb{R}_{>0} \times \mathbb{R}_{>0}$

$$\cdot \left| \frac{\partial(u, v)}{\partial(x, y)} \right| = \left| \det \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{bmatrix} \right|$$

$$= \left| \det \begin{bmatrix} y & x \\ -y/x^2 & 1/x \end{bmatrix} \right|$$

$$= \left| \frac{y}{x} - \left(-\frac{y}{x}\right) \right|$$

$$= \frac{2y}{x} \quad (\text{xy in first quadrant, so remove absolute})$$

Change of Variables in Multivariable Integration — Example 1 (cont'd)

$$\iint_D (x^2 + y^2) dA(x, y) = \iint_D \left(\frac{x^2 + y^2}{\left(\frac{2y}{x}\right)} \right) \cdot \frac{2y}{x} dA(x, y)$$

$$\frac{\frac{u}{v} + uv}{\left(\frac{2y}{x}\right)} = \frac{\frac{u}{v} + uv}{2v} = \frac{1}{2} \left(\frac{u}{v^2} + u \right)$$

try to write this purely in terms of u and v .

Jacobian only taken into account

$$\iint_D (x^2 + y^2) dA(x, y) = \iint_{[1,4] \times [1,4]} \frac{1}{2} \left(\frac{u}{v^2} + u \right) dA(u, v)$$

$$= \frac{1}{2} \iint_{[1,4] \times [1,4]} \left(\frac{u}{v^2} + u \right) dA(u, v)$$

$$= \frac{1}{2} \int_1^4 \left[\int_1^4 \left(\frac{u}{v^2} + u \right) dv \right] du$$

$$= \frac{1}{2} \int_1^4 \left[-\frac{u}{v} + uv \right]_{v=1}^{v=4} du$$

$$= \frac{1}{2} \int_1^4 \left[\underbrace{\left(-\frac{u}{4} + 4u \right)}_{\frac{15u}{4}} - \underbrace{(-u + u)}_0 \right] du$$

Change of Variables in Multivariable Integration — Example 1 (cont'd)

$$= \frac{1}{2} \int_1^4 \frac{15u}{4} du$$

$$= \frac{15}{8} \int_1^4 u du$$

$$= \frac{15}{8} \left[\frac{u^2}{2} \right]_1^4$$

$$= \frac{15}{8} \left(8 - \frac{1}{2} \right)$$

$$= 14.0625$$

Change of Variables in Multivariable Integration — Example 2

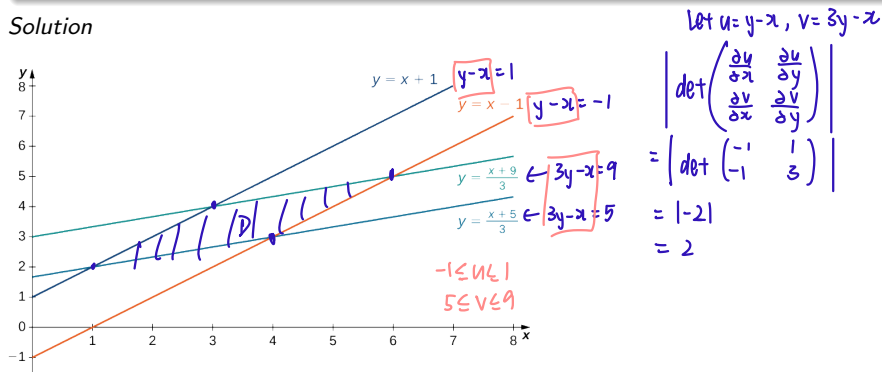
Example

Let R denote the closed parallelogram in $\mathbb{R}^2$ with vertices

$$(1, 2), \quad (3, 4), \quad (4, 3), \quad \text{and} \quad (6, 5).$$

Problem: Evaluate $\iint_D (x - y) \, dA(x, y)$.

Solution



Change of Variables in Multivariable Integration — Example 2 (cont'd)

$$\iint_D (x-y) \, dA(x,y) = \iint_D \frac{x-y}{2} \cdot 2 \, dA(x,y)$$

$$\frac{x-y}{2} = -\frac{u}{2}$$

$$\iint_D (x-y) \, dA(x,y) = \iint_{[-1,1] \times [5,9]} \boxed{\left(-\frac{u}{2}\right)} \, dA(u,v) \quad \begin{array}{l} \text{separable integral!} \\ \rightarrow -\frac{u}{2} = -\frac{u}{2} \cdot 1 \end{array}$$

$$= \int_{-1}^1 -\frac{u}{2} \, du \times \int_5^9 1 \, dv$$

$$= \left[-\frac{u^2}{4}\right]_{-1}^1 \cdot [v]_5^9$$

$$= 0$$

Change of Variables in Multivariable Integration — Example 2 (cont'd)

Change of Variables in Multivariable Integration — Example 3

Example

Let D denote the closed trapezoidal region in $\mathbb{R}^2$ with vertices

$$(1, 0), \quad (2, 0), \quad (0, -2), \quad \text{and} \quad (0, -1).$$

Problem: Evaluate $\iint_D e^{\frac{x+y}{x-y}} dA(x, y)$.

we try cos it's all in the exp

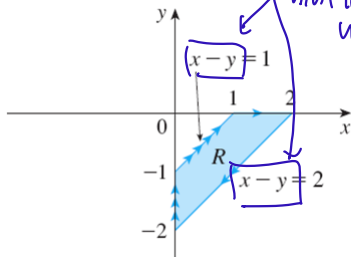
Solution

*hint let either
u or v = x-y*

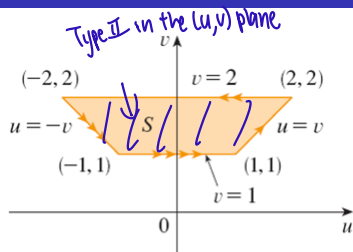
$$\text{let } u = x+y, \quad v = x-y$$

$$\left| \det \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} \right|$$

$$\begin{aligned} &= \left| \det \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \right| \\ &= |-2| \\ &= 2 \end{aligned}$$



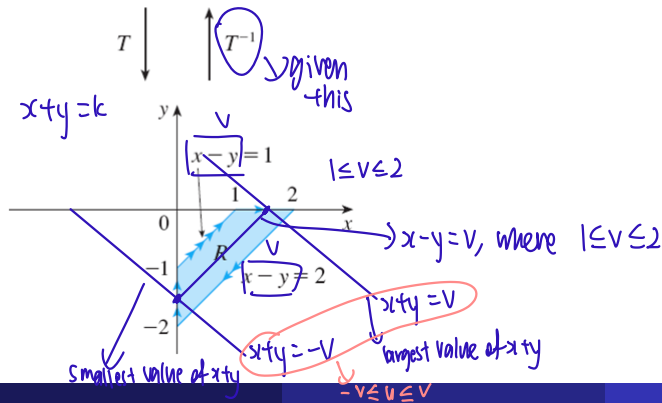
Change of Variables in Multivariable Integration — Example 3 (cont'd)



$$\iint_D e^{\frac{x+y}{x-y}} dA(x,y) = \iint_D \frac{e^{\frac{u}{v}}}{2} \cdot 2 dA(u,v)$$

$$\frac{e^{\frac{x+y}{x-y}}}{2} = \frac{1}{2} e^{\frac{u}{v}}$$

- $1 \leq v \leq 2$
- $u? -v \leq u \leq v$



Change of Variables in Multivariable Integration — Example 3 (cont'd)

$$\begin{aligned}
 \iint_D e^{\frac{x+y}{2}} dA(x,y) &= \int_1^2 \left[\int_{-v}^v \frac{1}{2} e^{\frac{u}{2}} du \right] dv && \text{Type II integral} \\
 &= \frac{1}{2} \int_1^2 \left[\int_{-v}^v e^{\frac{u}{2}} du \right] dv \\
 &= \frac{1}{2} \int_1^2 \left[ve^{\frac{u}{2}} \right]_{u=-v}^{u=v} dv && \text{chain rule} \\
 &= \frac{1}{2} \int_1^2 \left(ev - \frac{v}{e} \right) dv \\
 &= \frac{1}{2} \left(e - \frac{1}{e} \right) \cdot \int_1^2 v dv \\
 &= \frac{1}{2} \left(e - \frac{1}{e} \right) \cdot \left[\frac{v^2}{2} \right]_1^2 \\
 &= \frac{3}{4} \left(e - \frac{1}{e} \right)
 \end{aligned}$$

✱ Wouldn't have 3 variable for change of variables for exam.

Change of Variables in Multivariable Integration — Example 4

Example

Find the volume of the closed region in $\mathbb{R}^3$ bounded by the six planes given by

- $x + y + z = 1$ and $x + y + z = 2$; $1 \leq u \leq 2$
- $x + 2y = 0$ and $x + 2y = 1$; and $0 \leq v \leq 1$
- $y + z = 2$ and $y + z = 4$. $2 \leq w \leq 4$

Solution

$$\text{Let } u = x + y + z, \quad v = x + 2y, \quad w = y + z$$

$$\left| \det \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{pmatrix} \right| = \left| \det \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 0 \\ 0 & 1 & 1 \end{pmatrix} \right| = 2$$

$$\begin{aligned} \text{Volume} &= \iiint_R |dV(x, y, z)| \\ &= \iiint_R \frac{1}{2} \cdot 2 \, dV(x, y, z) = \iiint_{[1,2] \times [0,1] \times [2,4]} \frac{1}{2} \, dV(u, v, w) = 1 \end{aligned}$$

Let us look at this capybara chilling ...



Line integrals were invented in the early 19th century in order to solve urgent problems in physics involving the following concepts:

- Fluid flow.
- Forces.
- Electricity.
- Magnetism.
- Thermodynamics.

Line Integrals

We will learn how to evaluate the following line integrals:

- 1 The line integral of a scalar field f along a curve within $\text{Dom}(f)$.
- 2 The line integral of a vector field $\mathbf{F}$ along an oriented curve within $\text{Dom}(\mathbf{F})$.

Remarks:

- Moving forward, we make the blanket assumption that a curve in Euclidean space has a parametrization with the following two properties:
 - The parameter space is a bounded and closed interval I in $\mathbb{R}$. *curve has endpoints*
 - No two parameters yield the same point of the curve, with the possible exception of the endpoint parameters. *curve does not self-intersect*

We shall call such a parametrization “**nice**”.

- An **orientation** of a curve informally means a direction of travel along it.
- Every curve has precisely two orientations (under the blanket assumption).



Line Integrals — Scalar Fields

Let f be a real-valued function of several real variables.

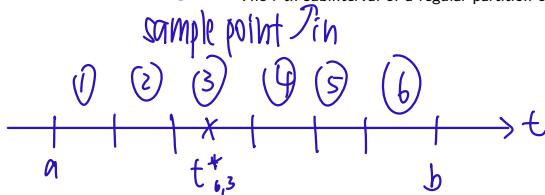
Let C be a curve contained within $\text{Dom}(f)$.

Choose a nice parametrization $\mathbf{r}$ of C , with parameter space $I = [a, b]$.

Recall that a sampling of I is a function t^* with the following two properties:

- $\text{Dom}(t^*) = \{(n, i) \in \mathbb{N}^2 \mid 1 \leq i \leq n\}$.
- $(n, i) \in \text{Dom}(t^*) \implies t_{n,i}^* \in \left[a + (i-1) \left(\frac{b-a}{n} \right), a + i \left(\frac{b-a}{n} \right) \right]$.

The i -th subinterval of a regular partition of I into n subintervals



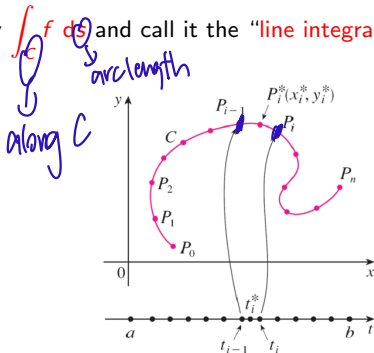
Line Integrals — Scalar Fields

If a real number L exists such that, regardless of the sampling t^* of I chosen,

$$L = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\mathbf{r}(t_{n,i}^*)) \cdot \underbrace{\text{dist}\left(\mathbf{r}\left(a + (i-1)\left(\frac{b-a}{n}\right)\right), \mathbf{r}\left(a + i\left(\frac{b-a}{n}\right)\right)\right)}_{(\Delta s)_i}, \quad \left. \begin{array}{l} \text{Not} \\ \text{tested} \end{array} \right\}$$

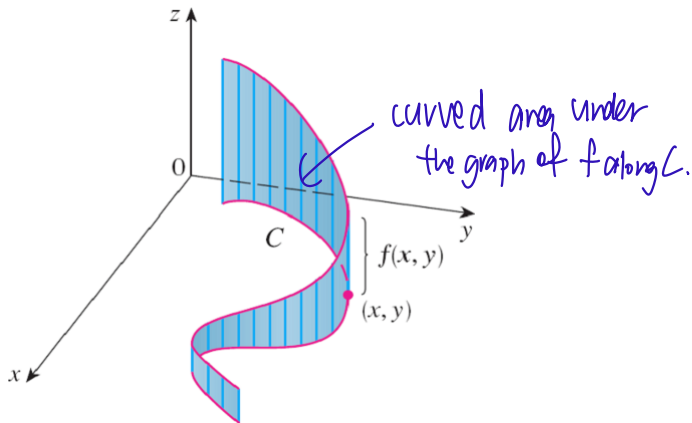
between 2 points

then we denote L by $\int_C f \, ds$ and call it the “line integral of f along C ”.



Remark: This definition does not depend on the nice parametrization $\mathbf{r}$ of C chosen.

Line Integrals — Scalar Fields



An informal geometric interpretation of $\int_C f \, ds$ is as follows: If f is non-negative, then

$\int_C f \, ds$ represents the "area" under the graph of f along the length of C .

↓
curved area

Line Integrals — Scalar Fields

Fractal curves typically have no well-defined length

Definition (Length of a Curve)

The **length** of a curve C , denoted by **Length(C)**, is defined to be $\int_C 1 \, ds$ — if it exists.

Good to know, not tested.

Theorem (Requirements for the Existence of Line Integrals of a Scalar Field)

- Let f be a continuous real-valued function of several real variables.
- Let C be a curve, contained within $\text{Dom}(f)$, whose length exists.

Then $\int_C f \, ds$ exists.

This theorem offers us complete assurance that in nearly all line-integral problems that we will encounter in this course, the line integral of a scalar field along a curve exists.

We now need a theorem to tell us how to actually compute the line integral of a scalar field along a curve.

Line Integrals — Scalar Fields

Definition (Smooth Curves and Smooth Nice Parametrizations)

A curve C in Euclidean space is said to be **smooth** if and only if we can choose for C a nice parametrization $\mathbf{r}$, with parameter space $I = [a, b]$, with the following properties:

- $\mathbf{r}'$ is continuous throughout $[a, b]$. $\vec{r}'$ is the tangent vector function.

- $\mathbf{r}'$ is non-zero at all parameters except, perhaps, for a and b .

$\vec{r}'(t) \neq \text{zero vector}$
except perhaps for a and b .

We shall call such a nice parametrization "**smooth**".

tested!

Theorem (Line Integrals of a Continuous Scalar Field Along a Smooth Curve)

- Let f be a continuous real-valued function of several real variables.
- Let C be a smooth curve contained within $\text{Dom}(f)$.

Then, for every smooth nice parametrization $\mathbf{r}$ of C , with parameter space $I = [a, b]$,

$$\int_C f \, ds = \int_a^b f(\mathbf{r}(t)) \cdot \|\mathbf{r}'(t)\| \, dt.$$

} how to compute $\int_C f \, ds$

Example

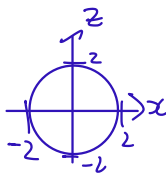
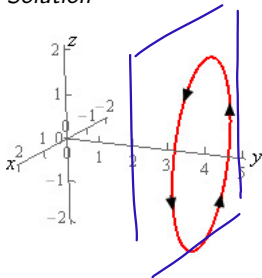
Define a continuous function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ by

$$f(x, y, z) \stackrel{\text{df}}{=} x^2 y^2.$$

Let C denote the circle in $\mathbb{R}^3$ in the $(y = 4)$ -plane with center $(0, 4, 0)$ and radius 2.

Problem: Evaluate $\int_C f \, ds$.

Solution



Project C onto the xz -plane

C is parameterized by the vector function $\vec{r} : [0, 2\pi] \rightarrow \mathbb{R}^3$, defined by $\vec{r}(t) = (2\cos(t), 4, 2\sin(t))$.

Line Integrals — Example 1 (cont'd)

For all $t \in [0, 2\pi]$:

$$\vec{r}'(t) = (-2\sin(t), 0, 2\cos(t)),$$

$$\begin{aligned} \|\vec{r}'(t)\| &= \sqrt{(-2\sin(t))^2 + 0^2 + (2\cos(t))^2} \\ &= \sqrt{4} = 2 \end{aligned}$$

$$\begin{aligned} \int_C f \, ds &= \int_0^{2\pi} f(\vec{r}(t)) \cdot \|\vec{r}'(t)\| \, dt \\ &= \int_0^{2\pi} f(2\cos(t), 4, 2\sin(t)) \cdot 2 \, dt \\ &= \int_0^{2\pi} [2\cos(t)]^2 \cdot 4^2 \cdot 2 \, dt \\ &= \int_0^{2\pi} 128 \cos^2(t) \, dt \\ &= 128 \int_0^{2\pi} \cos^2(t) \, dt \quad \begin{array}{l} \text{double} \\ \text{angle} \\ \text{formula} \end{array} \\ &= 128 \left[t + \frac{1}{2} \sin(2t) \right]_0^{2\pi} \\ &= 128\pi \end{aligned}$$

Definition (Vector Fields)

Let $n \in \mathbb{N}$.

An n -dimensional **vector field** is a function $\mathbf{F}$ with the following two properties:

- $\text{Dom}(\mathbf{F}) \subseteq \mathbb{R}^n$
- $\text{Range}(\mathbf{F}) \subseteq \mathbb{R}^n$.



We say that $\mathbf{F}$ is **smooth** if and only if its n scalar component functions, denoted by

$$F_1, \dots, F_n,$$

have continuous first-order partial derivatives defined throughout $\text{Dom}(\mathbf{F})$.

We call a 2-dimensional vector field a “**plane vector field**”.

We call a 3-dimensional vector field a “**space vector field**”.

auto
matic
in
ms-
power
point

Line Integrals — Vector Fields

Let $n \in \mathbb{N}$, and let $\mathbf{F}$ be an n -dimensional vector field.

Let C be an oriented curve contained within $\text{Dom}(\mathbf{F})$.

Choose a nice parametrization $\mathbf{r}$ of C — with parameter space $I = [a, b]$ — that yields the orientation of C .

If a real number L exists such that, regardless of the sampling t^* of I chosen,

$$L = \lim_{n \rightarrow \infty} \sum_{i=1}^n \mathbf{F}(\mathbf{r}(t_{n,i}^*)) \bullet \underbrace{\left[\mathbf{r}\left(a + (i-1)\left(\frac{b-a}{n}\right)\right) - \mathbf{r}\left(a + i\left(\frac{b-a}{n}\right)\right) \right]}_{(\Delta \mathbf{r})_i},$$

Not tested

Dot product
cos vector
not a dist. anymore. it's a vector

then we denote L by $\int_C \mathbf{F} \bullet d\mathbf{r}$ and call it the "line integral of $\mathbf{F}$ along C ".

Line Integrals — Vector Fields $\rightarrow$ depends on orientation!

From the preceding definition, the following statements can be shown:

- Given any two nice parametrizations, $\mathbf{r}$ and $\mathbf{s}$, of C that yield the orientation of C , calculating $\int_C \mathbf{F} \bullet d\mathbf{r}$ using $\mathbf{r}$ produces the same outcome as calculating it using $\mathbf{s}$.
- If one attempts to calculate $\int_C \mathbf{F} \bullet d\mathbf{r}$ using a nice parametrization that yields the opposite orientation of C , then $-\int_C \mathbf{F} \bullet d\mathbf{r}$ is obtained instead.

Theorem (Line Integrals of a Vector Field Are Affected by Curve Orientation)

- Let $n \in \mathbb{N}$, and let $\mathbf{F}$ be an n -dimensional vector field.
- Let C be an oriented curve contained within $\text{Dom}(\mathbf{F})$.
- Let $-C$ denote the reverse of C , i.e., C but with its orientation reversed.

Then $\int_C \mathbf{F} \bullet d\mathbf{r}$ exists if and only if $\int_{-C} \mathbf{F} \bullet d\mathbf{r}$ exists, in which case

$$\int_{-C} \mathbf{F} \bullet d\mathbf{r} = - \int_C \mathbf{F} \bullet d\mathbf{r}.$$

Line Integrals — Vector Fields

Theorem (Requirements for the Existence of Line Integrals of a Vector Field)

- Let $n \in \mathbb{N}$, and let $\mathbf{F}$ be a continuous n -dimensional vector field.
- Let C be a curve, contained within $\text{Dom}(\mathbf{F})$, whose length exists.

Then $\int_C \mathbf{F} \bullet d\mathbf{r}$ exists.

Good to know, not tested.

Tested!

Theorem (Line Integrals of a Continuous Vector Field Along a Smooth Curve)

- Let $n \in \mathbb{N}$.
- Let $\mathbf{F}$ be a continuous n -dimensional vector field.
- Let C be a smooth oriented curve contained within $\text{Dom}(\mathbf{F})$.

Then, for every smooth nice parametrization $\mathbf{r}$ of C — with parameter space $I = [a, b]$ — that yields the orientation of C ,

$$\int_C \mathbf{F} \bullet d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \bullet \mathbf{r}'(t) dt.$$

Diagram annotations:
- A box around the integrand $\mathbf{F}(\mathbf{r}(t)) \bullet \mathbf{r}'(t)$ is labeled "dot product".
- An arrow points from the word "vector" above $\mathbf{F}(\mathbf{r}(t))$ to the vector field symbol $\mathbf{F}$.
- An arrow points from the word "vector" above $\mathbf{r}'(t)$ to the derivative symbol $\mathbf{r}'$.
- An arrow points from the word "the tangent vector!" to $\mathbf{r}'(t)$.
- An arrow points from the word "for computation" to the integral symbol $\int$.

Example

Define a continuous space vector field $\mathbf{F}$ on $\mathbb{R}^3$ by

$$\mathbf{F}(x, y, z) \stackrel{\text{df}}{=} (e^{2x}, z(y+1), z^3).$$

Let C denote the oriented curve in $\mathbb{R}^3$ with a parametrization $\mathbf{r} : [0, 2] \rightarrow \mathbb{R}^3$ defined by

$$\mathbf{r}(t) \stackrel{\text{df}}{=} (t^3, 1 - 3t, e^t).$$

Problem: Evaluate $\int_C \mathbf{F} \bullet d\mathbf{r}$.

Solution

For all $t \in [0, 2]$: $\mathbf{r}'(t) = (3t^2, -3, e^t)$.
 $\mathbf{F}(\mathbf{r}(t)) = \mathbf{F}(t^3, 1 - 3t, e^t)$
 $= (e^{2t^3}, e^t(1 - 3t), e^{3t})$

Take the dot product!

Line Integrals — Example 2 (cont'd)

$$\begin{aligned}\int_C \vec{F} \cdot d\vec{r} &= \int_0^2 (e^{2t^3}, e^t(2-3t), e^{3t}) \cdot (3t^2, -3, e^t) dt \\&= \int_0^2 [3t^2 e^{2t^3} - 3e^t(2-3t) + e^{4t}] dt \quad \left. \begin{array}{l} \text{may have to use} \\ \text{integration by parts} \\ \text{and u-substitution} \end{array} \right\} \\&= \left[\frac{1}{2} e^{2t^3} - 3e^t(5-3t) + \frac{1}{4} e^{4t} \right]_0^2 \\&= \frac{57}{4} + 3e^2 + \frac{1}{4}e^8 + \frac{1}{2}e^6\end{aligned}$$